

Continuous attraction toward phonological competitors

Michael J. Spivey^{*†}, Marc Grosjean^{*‡§}, and Günther Knoblich^{*¶}

^{*}Department of Psychology, Cornell University, Ithaca, NY 14853; [†]Department of Psychology, Max Planck Institute for Human Cognitive and Brain Sciences, Amalienstrasse 33, Munich 80799 Germany; and [‡]Department of Psychology, Rutgers, The State University of New Jersey, Newark, NJ 07102

Communicated by James L. McClelland, Carnegie Mellon University, Pittsburgh, PA, May 10, 2005 (received for review January 28, 2005)

Certain models of spoken-language processing, like those for many other perceptual and cognitive processes, posit continuous uptake of sensory input and dynamic competition between simultaneously active representations. Here, we provide compelling evidence for this continuity assumption by using a continuous response, hand movements, to track the temporal dynamics of lexical activations during real-time spoken-word recognition in a visual context. By recording the streaming *x, y* coordinates of continuous goal-directed hand movement in a spoken-language task, online accrual of acoustic-phonetic input and competition between partially active lexical representations are revealed in the shape of the movement trajectories. This hand-movement paradigm allows one to project the internal processing of spoken-word recognition onto a two-dimensional layout of continuous motor output, providing a concrete visualization of the attractor dynamics involved in language processing.

dynamical systems | psycholinguistics | word recognition

Modular stage-based accounts of language processing have generally assumed that, rather than continuously cascading partial results of information processing to later stages (1, 2), the neural subsystems responsible for perception and cognition each wait until a stable unique representation has been computed before that information is passed on to the next processing stage (3–5). This kind of discrete stage-based theoretical framework has motivated much of cognitive psychology since its inception, and it continues to be a guiding force in various contemporary theories of cognitive processing. However, in the case of spoken-word recognition, a number of judgment-based experimental techniques have provided indirect evidence for partial activation of multiple lexical representations (“cohorts”) cascading to later stages of processing even just part of the way through hearing a word (6, 7). Moreover, recent eye-movement data have supported a continuously dynamic and highly interactive account of the real-time integration of information sources during spoken-language processing (8, 9). For example, eye movements to objects with phonologically similar names (e.g., saccades to a *candle* when instructed to “pick up the *candy*”) have been interpreted as evidence for continuous processing of phonological input and parallel activation of temporarily consistent lexical representations in monolingual adults (10, 11), bilingual adults (12), and children (13). Thus, it appears that neural patterns corresponding to multiple lexical representations may signal later stages of processing before the single correct lexical item is identified. However, it is still not entirely clear whether the activations of these lexical representations are updated continuously by the acoustic-phonetic input and constantly cascaded to later stages or whether there are intermediate noncascading stages in spoken-language comprehension.

Completely ruling out discrete-time incremental versions of the modular stage-based account (4) has proven to be difficult because there is, in principle, the possibility that the apparent continuity in recent results may be an artifact of averaging discontinuous or semicontinuous motor outputs (such as button

presses and saccades). In this work, we recorded a continuous hand-movement response during comprehension of spoken instructions in a visual context, and we show that it provides an unusually high-fidelity emission of the continuous cognitive dynamics inherent in real-time spoken-language processing.

Several computational models of spoken-word recognition assume relatively continuous input and parallel partial activation of lexical representations (10, 14–17). Corresponding to simulation results from the interactive-activation TRACE model of speech processing (14), the eye-movement data typically show a nonlinear rising curve over time for the probability of fixating the target object (referred to in the speech stream; e.g., “beaker”), and a significant rising-then-falling curve for the probability of fixating an object whose name has phonological overlap with the spoken word (e.g., a *beetle*, or a *speaker*) (18).

The semicontinuous record of eye position, alternating between steady fixations of 300–400 ms and fast, ballistic saccades of 20–40 ms, is a significant improvement over traditional outcome-based experimental methods that record only accuracy and reaction time at the end of a trial. Nonetheless, a disadvantage of the eye-movement evidence for parallel partial activation of lexical alternatives during spoken-word recognition is that it involves averaging “categorical” data (steady fixations of one object or another over time) to produce “continuous” functions. Thus, it can only approximate continuous central tendencies of group data.

Because saccades are largely ballistic (but cf. ref. 19), the experimental trials that contribute to evidence that the cohort lexical item is substantially active are always trials in which the participant briefly fixated directly on the cohort object at some point in the trial and then later fixated the target object before picking it up. In contrast to saccades, many arm movements are nonballistic and can often be smoothly redirected midflight (20). Therefore, by recording continuous arm movements in a similar visual display, one can observe graded effects of a competing object pulling the movement in its direction even on trials in which the hand only ever settles on the correct target object.

Experiment

Methods. Forty-two Cornell University undergraduates participated in the experiment for extra credit in psychology courses. Participants were presented with color images of two objects on a screen (one target and one distractor), and a prerecorded speech file instructed them to click one of them with the mouse. Objects were presented in the upper left and upper right corners of the computer screen (e.g., a *candle* and a *candy*, in the cohort condition, or a *candle* and a *jacket*, in the control condition). Eight target objects were used to make 32 trials in which the distractor object was either a cohort for the target object or a

See Commentary on page 9995.

[†]To whom correspondence may be addressed. E-mail: spivey@cornell.edu, grosjean@cbs.mpg.de, or knoblich@psychology.rutgers.edu.

[§]Present address: Institute for Occupational Physiology, University of Dortmund, Ardeystrasse 67, 44139 Dortmund, Germany.

© 2005 by The National Academy of Sciences of the USA

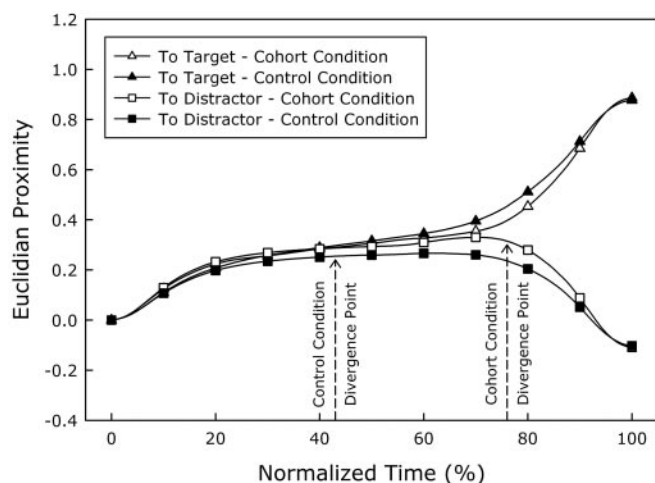


Fig. 2. Proximity of the mouse cursor to target (triangles) and to distractor (squares) over normalized time in the control condition (filled symbols) and in the cohort condition (open symbols). Averaged movement trajectories exhibit a significantly greater proximity to target than to distractor much earlier in the control condition than in the cohort condition.

distractor objects over time can be similarly treated as an indicator of the activation of the competing lexical representations. Fig. 2 shows the proportional Euclidean proximity, $1 - \text{distance}/\text{max}(\text{distance})$, to the centers of the target object and the distractor object over normalized time (averaged across leftward and rightward movements). During early portions of the movement trajectory, proximity to the target and distractor are not significantly different from one another, because the movement is largely in the vertical dimension. In the control condition, these two proximities diverge significantly from one another at the 43rd normalized time slice, and they continue to be significantly diverged for the remaining 57% of the movement duration. In the cohort condition, they do not significantly diverge until the 76th normalized time slice. Moreover, showing that the cohort object attracted the movement toward itself, proximity to the distractor in the cohort condition was significantly greater than proximity to the distractor in the control condition ($P < 0.05$) all of the way from the fourth to the 93rd normalized time slices. As evidence that the activation of the spoken lexical item may reach asymptote less quickly when an object with a phonologically competing name is visually present, proximity to the target object in the cohort condition was significantly lower than proximity to the target object in the control condition from the 66th to the 91st normalized time slices.

Recall that mouse movement was initiated, on average, at ≈ 345 ms after visual onset of the objects, and speech onset was exactly 500 ms after visual onset of the objects. Therefore, by the time the speech input began, participants were, on average, already ≈ 155 ms into their $\approx 1,400$ -ms mouse-movement trajectory. Therefore, the fact that the divergence between distractor proximity in the cohort and control conditions is observed as early as the fourth normalized time step (Fig. 2) indicates that in a proportion of trials in which movement initiation was later than average, there was sufficient distinguishing information in the spoken input to reliably sway the trajectory almost immediately.

To cement the claim that movement trajectories in the cohort condition are all statistically deflected toward the competitor object in a continuously graded fashion, it is necessary to look at the distribution of these deflections across multiple trials. In principle, it could be the case that, as with saccadic eye move-

ments, there are some trials in which the competitor object does not attract the motor output and other trials in which it does. Such a bimodal distribution could be consistent with a stage-based account of spoken-word recognition in which an incorrect interpretation of the spoken word is occasionally briefly instated in discrete symbolic form (thus triggering a motor output toward that competitor object), and then quickly replaced by the correct symbolic lexical form (thus triggering a corrective movement toward the object being referred to). When averaged, this hypothetical data pattern would produce mean movement trajectories that could falsely suggest simultaneous partial activation and competition among multiple lexical representations.

To examine this possibility, we calculated the degree of curvature (toward the distractor object) among the trajectories in the cohort and control trials in terms of the area (in pixels) between each actual trajectory and a straight line connecting its start and endpoint. (Portions of curvature away from the distractor object and away from the straight line resulted in negative area calculations.) With too few trials within a participant to provide an adequate measure of the unimodality or bimodality of the distribution of these trajectory deflections, the values for both cohort and control conditions were together converted into z scores within a participant and then pooled across participants. Fig. 3A shows the z distribution for the cohort trials ($n = 611$; mean, 0.164; variance, 1.043; kurtosis, 0.76; skewness, 0.658), looking quite similar to that of the control trials ($n = 606$; mean, -0.165 ; variance, 0.889; kurtosis, 1.75; skewness, 0.861). For the cohort z distribution, the bimodality coefficient (b) was 0.381, and for the control z distribution, it was 0.366 (with $b > 0.555$ being the standard cutoff for multimodality). Note that if continuous eye movement scan paths sampled at 60+ Hz (instead of fixation analyses) were subjected to corresponding curvature analyses, there would be a decidedly bimodal pattern in the distribution. Participants in those studies either fixate the competitor object or they do not, on any given trial. They do not make saccades slightly toward the competitor object the way these mouse-movement trajectories show deflections slightly toward the competitor object.

In a Kolmogorov–Smirnov test of normality, the cohort z distribution was not significantly different from a normal distribution with the same mean and variance, but high kurtosis (indicating an unusually high proportion of trials near the mean) did make the test marginally significant ($P = 0.054$). In such a test with the control z distribution, the even higher kurtosis caused it to be significantly different from a normal distribution with the same mean and variance ($P = 0.023$). In both cases, the deviation from normality is due to high kurtosis, meaning that the distributions are even more sharply singly peaked and further from bimodality than their corresponding normal distributions with matched mean and variance.

We also z -scored, within each participant, the area under the trajectory separately for cohort and control trials, and we then pooled across subjects (see Fig. 3B). With these two z distributions having the same mean (of zero) and the same variance (of 0.966), the Kolmogorov–Smirnov test can evaluate the difference between their respective shapes (e.g., skewness, kurtosis, and multimodality). With no theoretical reason to imbue bimodality in the control z distribution, quantitative evidence for high similarity between the control and cohort z distributions would substantially allay concerns that some hidden bimodal behavior exists in the cohort condition. When comparing these two z distributions, the Kolmogorov–Smirnov test produces a P value > 0.9999 .

In sum, three tests cast considerable doubt on the possibility of the cohort condition being composed of (*i*) some trials that behave like control trials (indicating no competing lexical activation of the cohort item) and (*ii*) some trials that exhibit uniquely curved trajectories (consistent with a discrete tempo-

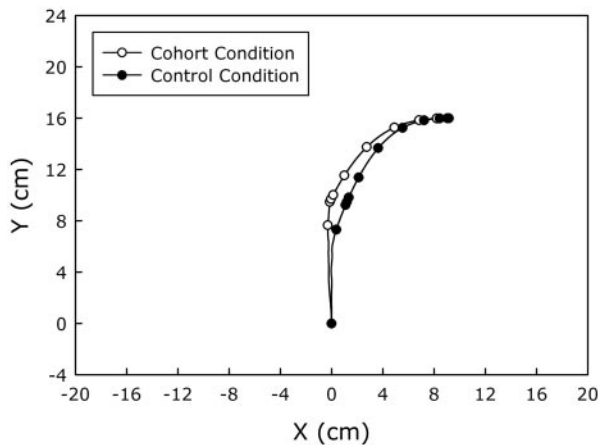


Fig. 4. Simulated movement trajectories resulting from interfacing the TRACE model of spoken-word recognition with a localist attractor network and stochastically sampling x and y mouse movements from its visual object nodes. When the cohort distractor object is present ($\circ$), the simulated movement trajectory gravitates to the region in between the two objects for longer than when the control distractor object is present ($\bullet$).

envelope over time, which approximated the observed U-shaped velocity profile in the human data, produced the slow down that occurs in both conditions around the middle of the movement.

$$\Delta \text{pos} = \Delta \text{pos} \cdot \left(1 - \frac{125}{50\sqrt{2\pi}} e^{-\frac{(t-50)^2}{2 \cdot 50^2}} \right) \quad [2]$$

Also, multiplying each x and y increment by the inverse proportion of the current distance from the x, y position of the goal in cm, $1 - x/10$, and $1 - y/16$, produced the slow down that occurs in both conditions as the goal is reached (see, for example, the smooth final approach to asymptote in Fig. 2). Last, for comparison with the human data, the 170 time steps of x, y coordinates of the model were normalized to 101 time slices.

Results. Fig. 4 plots the mean x, y movement trajectories (in cm) from 10 runs of the model in the cohort condition and the control condition. Much as in the human data (Fig. 1), when the lexical representation of the distractor object is initially accruing partial activation at a similar rate to the lexical representation of the target object (cohort condition), the simulated movement trajectory continues upward between, and equidistant from, the two objects for a longer period than when the name of the distractor object exhibits no phonological similarity to the spoken word (control condition).

When proximity to target and distractor over time is plotted from these simulated changes in x, y position (Fig. 5), the resulting pattern bears considerable similarity to the human data (Fig. 2), although the current simulation does exhibit divergence somewhat earlier than the human data. Comparing the four curves at every normalized time slice in this image to the four curves at every time slice in Fig. 2 produces a root-mean-squared error of 0.0625, and $r^2 = 0.76$ ($P < 0.0001$).

As for variability in trajectory curvature across multiple runs of the simulation, the distributions are, not surprisingly, highly normal. When the area between each trajectory and its straight line was calculated for 600 runs of the model in each condition and the data were z scored together, the z distribution for the cohort trials (mean, 0.114; variance, 0.997; kurtosis, -0.520 ; skewness, -0.027) was quite similar to that of the control trials (mean, -0.114 ; variance, 0.997; kurtosis, -0.488 ; skewness, 0.129). The distributions are shown in Fig. 6, bearing some resemblance to Fig. 3A. Their bimodality coefficients were $b =$

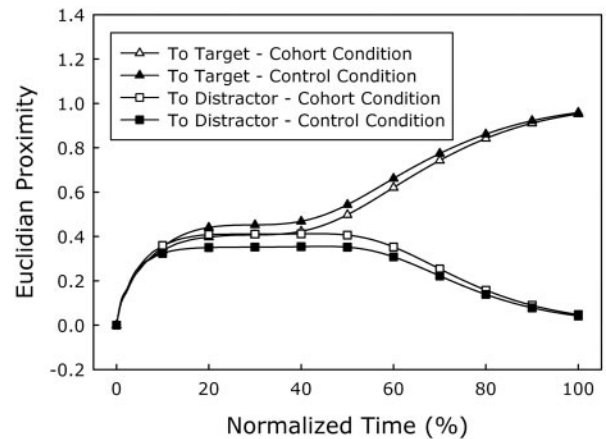


Fig. 5. Proximity to target (triangles) and to distractor (squares) over normalized time in the control condition (filled symbols) and the cohort condition (open symbols) from the simulated mouse movements triggered by graded lexical activation continuously spreading to visual object nodes.

0.403 and $b = 0.404$, respectively. Rather than exhibiting the unusually sharp peak near the mean that was seen in the human data, the distributions of the simulation conformed quite closely to a normal distribution. In the Kolmogorov–Smirnov normality test, both cohort and control z distributions did not remotely differ from their corresponding (matched mean and variance) normal distributions (both $P > 0.9999$). When this normality test was used to compare the cohort z distribution of the simulation with the cohort z distribution of the human data (because their means and variances are quite similar), the difference was not significant ($P = 0.142$). However, when the control z distribution of the simulation was compared with the human data control z distribution, they were significantly different ($P = 0.007$). The notable differences in kurtosis and skewness between the distributions of the human data and the simulation results remain to be examined.

General Discussion

These results provide powerful support for models of continuous uptake of acoustic–phonetic input during spoken-word recognition. The substantial fit between model simulation and human data provides an encouraging, if simplified, linking hypothesis to

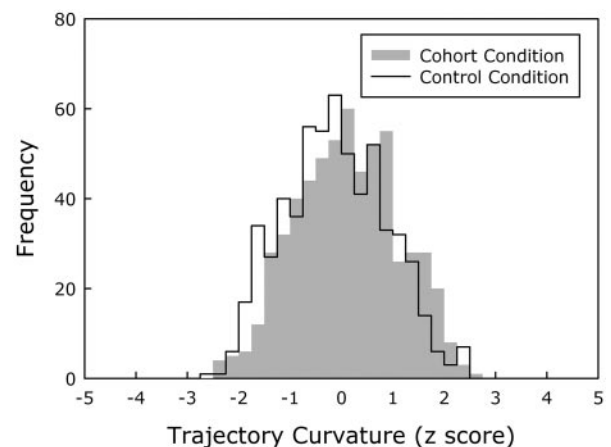


Fig. 6. Overlaid histograms of the trajectory curvature of the simulation in the cohort and control conditions (z scored together over all 1,200 runs) reveal normal unimodal distributions with slightly lower kurtosis and skewness than those from the human data shown in Fig. 3A.

